

Explainable Deep Learning for Breast Cancer Detection Using Ultrasound Imaging

Sarfo K. Frimpong

A thesis submitted to the Faculty of The College of Graduate Studies at Virginia State University in partial fulfillment of the requirements for the degree of Master of Science in the Department of Computer Science in the College of Engineering and Technology

Virginia

2026

Approved by

Chair/Advisor: Chen, Wei Bang

Date: April 24, 2026

Committee Member: Khounthun Xiang

Date: April 24th, 2026

Committee Member: 

Date: April 24th, 2026

Committee Member: 

Date: April 24th, 2026

Abstract

Sarfo K. Frimpong

Explainable Deep Learning for Breast Cancer Detection Using Ultrasound Imaging.

(Under the direction of Dr. Wei-Bang Chen, Dr. Xuanchen Xiang, Dr. Cesa Salaam, Dr. Joseph A. Shelton,
and Dr. Qian Qu)

This study presents an explainable deep learning framework for automated malignant detection in breast ultrasound imaging. Using the Breast Ultrasound Dataset (BUSI) (Al-Dhabyani et al., 2020), a transfer learning-based ResNet-18 model (He et al., 2016) was fine-tuned for binary classification of malignant versus non-malignant cases. On a held-out test set ($n = 117$), the model achieved 94.87% accuracy, 93.10% precision, 87.10% sensitivity, and 97.7% specificity, with an area under the receiver operating characteristic curve (ROC-AUC) of 0.9816, indicating strong class separability. This study adopts a specificity-driven evaluation framework, prioritizing the reduction of false positives to minimize unnecessary biopsies, patient anxiety, and healthcare costs. While maintaining clinically acceptable sensitivity, the model demonstrates strong capability in correctly identifying non-malignant cases. To enhance interpretability, Gradient-weighted Class Activation Mapping (Grad-CAM) (Selvaraju et al., 2017) was applied, and activation maps were qualitatively compared with expert-annotated lesion masks. Results show consistent localization of diagnostically relevant regions in correctly classified cases. The integration of transfer learning, specificity-oriented evaluation, and explainable AI contributes toward clinically reliable and interpretable decision-support systems for breast cancer detection.

Keywords— Breast Cancer, Deep Learning, ResNet-18, Grad-CAM, Explainable AI.

Table of Contents

List of Tables	iii
List of Figures	iv
CHAPTER 1 – INTRODUCTION	1
1.01 Introduction.....	1
1.02 Artificial Intelligence in Breast Cancer Detection.....	1
1.03 Problem Definition.....	1
1.04 Data Preprocessing.....	1
1.05 Problem Formulation	2
1.06 Dataset Split.....	2
1.07 Model Performance.....	3
1.08 Explainability and Clinical Alignment.....	3
1.09 Research Aim and Contributions	4
CHAPTER 2 – ANALYTICAL AND REVIEW FOUNDATION	6
2.01 Deep Learning in Breast Ultrasound Classification.....	6
2.02 Residual Networks and Transfer Learning	6
2.03 Clinical Evaluation in Diagnostic Context	7
2.04 Explainable Artificial Intelligence in Oncology	8
2.05 Qualitative Lesion Localization and Research Positioning	9
CHAPTER 3 – RESEARCH FRAMEWORK AND EXPERIMENTAL DESIGN	11

3.01	Research Problems	11
3.02	Dataset and Experimental Setting.....	12
3.03	Model Architecture and Training	12
3.04	Model Explainability	13
3.05	Research Questions	14
3.06	Research Hypotheses	14
CHAPTER 4 – RESULTS AND DISCUSSION.....		15
4.01	Training Performance Analysis.....	15
4.02	Performance Metrics	15
4.03	Confusion Matrix Analysis	15
4.04	Analysis of Misclassification.....	16
4.05	Imaging Limitations and Clinical Implications	16
4.06	Explainability and Clinical Alignment.....	16
4.07	Summary of Findings.....	17
CHAPTER 5 – CONCLUSION AND FUTURE WORK.....		19
5.01	Research Conclusions	19
5.02	Limitations	19
5.03	Future Work	19
5.04	Final Remarks	19
References		19

List of Tables

Table 1. Dataset Distribution..... 12

Table 2. Performance Metrics for Breast Cancer Classification 15

Table 3. Confusion Matrix for Malignant vs Non-Malignant Classification 15

List of Figures

Figure 1. Confusion Matrix.....	3
Figure 2. Training Accuracy Curve.....	7
Figure 3. Correctly classified malignant case with Grad-CAM visualization (True = 1, Predicted = 1, P(mal) = 1.00).	9

CHAPTER 1 – INTRODUCTION

1.01 Introduction

Cancer remains one of the leading causes of mortality worldwide, with breast cancer representing one of the most prevalent malignancies affecting women (World Health Organization [WHO], 2023; American Cancer Society, 2023). Early detection is the most critical factor influencing survival outcomes. When identified at early stages, five-year survival rates exceed 90%, whereas late-stage diagnosis significantly reduces prognosis and increases treatment complexity (American Cancer Society, 2023; National Cancer Institute [NCI], 2023).

Breast ultrasound imaging plays an essential role in diagnostic workflows, particularly for patients with dense breast tissue where mammography sensitivity may be limited (Skaane, 2017; Kopans, 2007). Ultrasound is non-ionizing, cost-effective, and widely accessible, making it an important complementary modality in breast cancer screening and diagnostic evaluation (Kopans, 2007). However, interpretation remains highly dependent on radiologist expertise and is subject to inter-observer variability caused by operator skill, image quality, and diagnostic fatigue (Doi, 2007).

As screening volumes increase globally, healthcare systems face growing demand for consistent and scalable diagnostic support mechanisms. These challenges motivate the development of artificial intelligence (AI)-based systems capable of augmenting radiological assessment, improving diagnostic reliability, and reducing human variability in interpretation (Litjens et al., 2017).

1.02 Artificial Intelligence in Breast Cancer Detection

Artificial intelligence (AI), particularly deep learning, has demonstrated significant promise in medical image analysis (Litjens et al., 2017). Convolutional neural networks (CNNs) automatically learn hierarchical feature representations from raw pixel data, eliminating the need for handcrafted feature engineering.

In oncology, CNN-based models have achieved strong performance across multiple imaging modalities, including mammography, computed tomography (CT), dermoscopy, and ultrasound imaging (Esteva et al., 2017; Rajpurkar et al., 2018). Large-scale studies have shown that deep learning systems can approach or exceed specialist-level performance in selected diagnostic tasks.

However, predictive accuracy alone is insufficient for clinical deployment. AI systems must demonstrate reliability, transparency, and clinically meaningful evaluation to ensure safe and responsible integration into healthcare workflows (Doi, 2007; Parikh et al., 2019).

1.03 Problem Definition

This study investigates automated detection of malignant lesions in breast ultrasound images using the Breast Ultrasound Dataset (BUSI), which contains benign, malignant, and normal cases (Al-Dhabyani et al., 2020). The goal is to develop a deep learning model that accurately distinguishes malignant from non-malignant cases while maintaining interpretability.

The experimental framework was implemented using a PyTorch-based pipeline incorporating preprocessing, transfer learning with ResNet-18, and Grad-CAM for explainability. The full implementation is available in the publicly accessible repository: “BUSI Explainable CNN with Grad-CAM” (Frimpong, 2026), ensuring reproducibility and alignment between the methodology and implementation.

1.04 Data Preprocessing

All images were resized to 224×224 pixels to match the input requirements of the ResNet-18 architecture. Pixel values were normalized using a mean of 0.5 and a standard deviation of 0.5, resulting in an approximate range of $[-1, 1]$, which improves training stability and convergence.

Data augmentation techniques, including random horizontal flipping and rotation, were applied exclusively to the training set to improve generalization. Validation and test sets were not augmented to ensure unbiased evaluation.

Although contrast enhancement techniques such as histogram equalization and Contrast Limited Adaptive Histogram Equalization (CLAHE) are commonly used in medical imaging, they were not incorporated into the final model in order to preserve the original clinical characteristics of the ultrasound images.

To ensure reproducibility and methodological consistency, all preprocessing steps described in this study were directly implemented in the experimental pipeline. Specifically, normalization was performed using mean = 0.5 and standard deviation = 0.5, and augmentation was limited to horizontal flipping and rotation. No additional enhancement techniques were applied in the final model.

ResNet-18 was selected due to its balance between model complexity and computational efficiency. Its residual learning framework enables effective training of deeper representations while mitigating vanishing gradient issues (He et al., 2016). Additionally, pretrained ImageNet weights support transfer learning, improving generalization performance on limited medical imaging datasets (Litjens et al., 2017).

1.05 Problem Formulation

For binary classification, malignant cases were encoded as 1, while benign and normal cases were grouped together and encoded as 0.

After binary conversion, this resulted in:

- 570 negative samples
- 210 positive samples

1.06 Dataset Split

The dataset was split into training, validation, and test subsets. The held-out test set contained 117 images:

- 86 negative
- 31 malignant

1.07 Model Performance

On the held-out test set, the fine-tuned ResNet-18 model achieved an overall accuracy of 94.87%, with a precision of 93.10%, sensitivity (recall) of 87.10%, specificity of 97.7%, an F1-score of 0.90, and a receiver operating characteristic area under the curve (ROC-AUC) of 0.9816.

The confusion matrix is structured based on actual versus predicted class labels. The model correctly identified 84 non-malignant cases (true negatives) and 27 malignant cases (true positives). Only 2 false positives and 4 false negatives were observed, demonstrating strong discriminative capability. The low number of false positives directly contributes to the high specificity of the model.

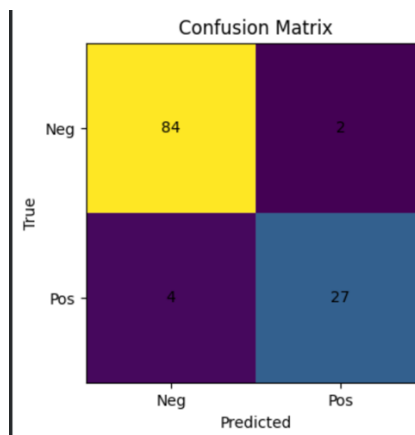


Figure 1. Confusion Matrix

1.08 Explainability and Clinical Alignment

To improve interpretability, Gradient-weighted Class Activation Mapping (Grad-CAM) was applied to generate class-discriminative localization maps that highlight regions influencing model predictions (Selvaraju et al., 2017).

Grad-CAM visualizations were qualitatively analyzed by overlaying activation maps on ultrasound images. In correctly classified malignant cases, activation regions were concentrated over lesion areas and showed visual

correspondence with expert-annotated masks from the BUSI dataset (Al-Dhabyani et al., 2020). Misclassified cases exhibited more diffuse or spatially shifted attention patterns.

This qualitative analysis provides insight into model behavior and supports interpretability without relying on purely black-box predictions.

1.09 Research Aim and Contributions

The objective of this thesis is to develop an explainable deep learning framework for breast ultrasound cancer detection that balances predictive performance with interpretability.

This study adopts a specificity-driven evaluation framework aligned with diagnostic requirements. The detailed justification for this evaluation approach is provided in Chapter 2. The contributions of this work include:

i. Transfer Learning Framework:

A fine-tuned ResNet-18 model is employed for binary classification, leveraging pretrained features to improve performance and optimization stability (He et al., 2016).

ii. Specificity-Driven Evaluation

Model performance is evaluated with emphasis on specificity to reduce false positive predictions.

Additional metrics, including accuracy, precision, and ROC-AUC, are also reported.

iii. Explainability Integration:

Grad-CAM is applied to generate visual explanations of model predictions, enhancing transparency and trust in AI-assisted diagnosis (Selvaraju et al., 2017).

iv. Qualitative Lesion Localization Analysis:

Activation maps are qualitatively compared with expert-annotated lesion masks to assess spatial correspondence with clinically relevant regions.

This work emphasizes reproducibility, robustness, and clinical relevance, contributing toward the development of reliable and interpretable AI-assisted diagnostic systems.

CHAPTER 2 – ANALYTICAL AND REVIEW FOUNDATION

2.01 Deep Learning in Breast Ultrasound Classification

Deep learning has significantly transformed medical image analysis. In breast cancer diagnostics, convolutional neural networks (CNNs) have demonstrated strong performance across mammography, ultrasound, and MRI modalities. Unlike traditional machine learning approaches that rely on handcrafted features, CNNs learn hierarchical representations directly from image data through end-to-end optimization (Litjens et al., 2017).

Breast ultrasound imaging presents unique challenges for automated classification. Images are often characterized by speckle noise, low contrast between lesions and surrounding tissue, irregular lesion boundaries, and operator-dependent variability. These factors increase intra-class variability and reduce signal consistency, making ultrasound classification more complex than other imaging modalities.

Although CNN models commonly report accuracy between 85% and 95%, results vary across datasets due to differences in preprocessing, class imbalance, and evaluation protocols. Many studies emphasize overall accuracy, which may obscure clinically relevant performance characteristics.

2.02 Residual Networks and Transfer Learning

Residual Networks (ResNets) introduced shortcut connections that improve gradient flow and enable stable training of deeper architectures. The residual formulation is defined as:

$$H(x) = F(x) + x$$

where the network learns a residual function $F(x)$ that is added to the input x , improving convergence and mitigating vanishing gradient issues (He et al., 2016).

ResNet-18 was selected in this study due to its balance between representational capacity and computational efficiency. Fine-tuning achieved high training and validation performance, indicating effective feature learning and generalization to ultrasound-specific characteristics.

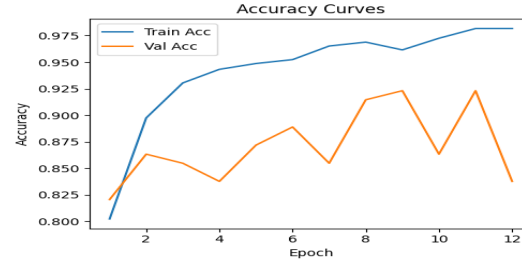


Figure 2. Training Accuracy Curve

2.03 Clinical Evaluation in Diagnostic Context

In diagnostic workflows, evaluation metrics must reflect clinical decision-making priorities rather than broad detection objectives. Unlike screening scenarios, which aim to identify as many potential cases as possible, diagnostic systems are used to support clinical decisions regarding further intervention.

In this context, specificity is a critical evaluation metric, as false positive predictions can lead to unnecessary biopsies, invasive procedures, increased healthcare costs, and patient anxiety (Doi, 2007; Parikh et al., 2019). Therefore, minimizing false positives is essential for improving diagnostic reliability and reducing clinical burden.

Specificity is defined as:

$$\frac{TN}{TN + FP}$$

where true negatives (TN) represent correctly identified non-malignant cases, and false positives (FP) represent benign cases incorrectly classified as malignant.

In this study, initial experiments achieved a specificity of approximately 94%. After model optimization, specificity improved to 97.7%, with only two false positive predictions. This improvement demonstrates enhanced

discrimination between malignant and non-malignant cases and supports the suitability of the model for diagnostic decision-support applications.

While multiple evaluation metrics are reported, this study prioritizes specificity to align with the requirements of diagnostic use.

2.04 Explainable Artificial Intelligence in Oncology

Deep learning models are often criticized for limited interpretability. In clinical environments, transparency is essential because diagnostic decisions directly influence treatment planning and patient outcomes.

Gradient-weighted Class Activation Mapping (Grad-CAM) provides visual explanations by generating class-discriminative localization maps based on feature importance (Selvaraju et al., 2017). These maps highlight regions of the image that contribute most to model predictions, enabling clinicians to better understand model behavior.

In this study, Grad-CAM is applied to the final convolutional layer of ResNet-18 and overlaid on ultrasound images to visualize model attention and support interpretability.

Grad-CAM generates class-discriminative localization maps:

$$L_{Grad-CAM}^c = ReLu\left(\sum_K a_k^c A^k\right)$$

where feature maps are weighted by class-specific gradients.

2.05 Qualitative Lesion Localization and Research Positioning

A key advantage of the Breast Ultrasound Dataset (BUSI) is the availability of expert-annotated lesion masks (Al-Dhabyani et al., 2020). While many studies use visualization techniques, fewer assess whether model attention aligns with clinically relevant structures (Litjens et al., 2017; Selvaraju et al., 2017).

In this study, Grad-CAM heatmaps are qualitatively compared with ground-truth masks. Correctly classified malignant cases show concentrated activation over lesion regions, as illustrated in Figure 3, indicating that the model focuses on diagnostically relevant structures.

This qualitative analysis supports interpretability by demonstrating that predictions are based on meaningful image regions rather than background artifacts. By integrating classification performance with visual explanation, this study contributes to the development of clinically interpretable AI systems.

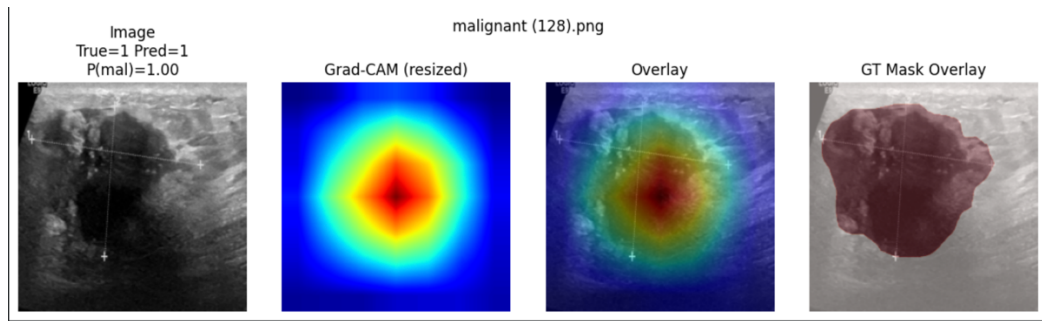


Figure 3. Correctly classified malignant case with Grad-CAM visualization ($True = 1$, $Predicted = 1$, $P(mal) = 1.00$).

This qualitative assessment provides insight into model behavior without relying on quantitative localization metrics. The integration of classification performance with visual explanation addresses key gaps in existing literature, including:

- i. Overreliance on overall accuracy without a clinically meaningful evaluation
- ii. Limited integration of explainability into model assessment
- iii. Lack of analysis of model attention relative to pathological regions

In diagnostic settings, evaluation must reflect clinical priorities rather than relying solely on aggregate accuracy. In breast cancer diagnosis, specificity is a critical metric, as false positive predictions can lead to unnecessary biopsies, increased healthcare costs, and patient anxiety (Doi, 2007; Parikh et al., 2019). Specificity is defined as:

$$\frac{TN}{TN + FP}$$

where true negatives (TN) represent correctly identified non-malignant cases, and false positives (FP) represent benign cases incorrectly classified as malignant.

Initial experiments achieved a specificity of approximately 94%. After optimization, specificity improved to 97.7%, with only two false positive predictions. This improvement demonstrates enhanced discrimination between malignant and non-malignant cases and highlights the effectiveness of the proposed framework in reducing unnecessary clinical interventions.

Although multiple metrics are reported, this study prioritizes specificity to align with clinical objectives in minimizing overdiagnosis. By combining transfer learning, specificity-driven evaluation in a diagnostic context, and explainable AI, this work contributes toward the development of clinically reliable and interpretable breast cancer detection systems.

CHAPTER 3 – RESEARCH FRAMEWORK AND EXPERIMENTAL DESIGN

3.01 Research Problems

Breast ultrasound imaging plays a critical role in breast cancer diagnosis, particularly for patients with dense breast tissue where mammography performance may be limited. Despite its clinical importance, ultrasound interpretation remains highly dependent on radiologist expertise and is affected by image noise, lesion variability, and operator-dependent acquisition. These factors introduce diagnostic variability and may result in misclassification or unnecessary clinical interventions (Doi, 2007).

Recent advances in artificial intelligence, particularly convolutional neural networks (CNNs), have demonstrated strong capability in medical image classification tasks (Litjens et al., 2017). However, several challenges remain before such systems can be reliably integrated into clinical workflows. Many existing models emphasize overall accuracy without sufficient consideration of clinically relevant evaluation metrics. In diagnostic settings, reducing false positive predictions is especially important, as they can lead to unnecessary biopsies, increased healthcare costs, and patient anxiety (Parikh et al., 2019).

In addition, deep learning models often lack interpretability. Clinical adoption requires transparency to ensure that predictions are based on meaningful pathological features rather than spurious image artifacts. Without visual explanation mechanisms, clinicians cannot verify whether automated predictions align with anatomically relevant regions.

This research therefore investigates whether an explainable deep learning framework can achieve strong diagnostic performance while maintaining interpretability and clinical plausibility in breast ultrasound cancer detection.

3.02 Dataset and Experimental Setting.

The experiments were conducted using the Breast Ultrasound Dataset (BUSI), which contains ultrasound images categorized into benign, malignant, and normal classes (Al-Dhabyani et al., 2020). To align the task with clinically meaningful malignant detection, the dataset was reformulated as a binary classification problem, where malignant cases were treated as positive samples and benign or normal cases as negative samples.

After preprocessing and dataset preparation, the final dataset distribution consisted of:

- 570 negative samples (benign + normal)
- 210 malignant samples

The dataset was divided into training, validation, and testing subsets:

Table 1. Dataset Distribution

Dataset split	Total Images	Negative	Positive
Training	546	399	147
Validation	117	85	32
Test	117	85	31

The held-out test set of 117 images was used exclusively for final performance evaluation

3.03 Model Architecture and Training

A transfer learning approach using ResNet-18 was employed. ResNet uses residual connections that allow deeper networks to train effectively by learning residual mappings:

$$H(x) = F(x) + x$$

The model was initialized with ImageNet pretrained weights, enabling transfer of low-level visual features such as edges and textures to the medical imaging task.

Training was conducted for **12 epochs**. Training accuracy gradually increased to approximately **98%**, while validation accuracy stabilized around **92–93%**. Training loss decreased consistently across epochs, while validation loss fluctuated slightly but remained stable, suggesting reasonable generalization with limited overfitting.

3.04 Model Explainability

To improve interpretability, Gradient-weighted Class Activation Mapping (Grad-CAM) was applied to visualize the regions of ultrasound images influencing model predictions (Selvaraju et al., 2017). Grad-CAM generates class-discriminative localization maps using gradients from the final convolutional layer, defined as:

$$L_C^{Grad-CAM} = ReLu \left(\sum_k \alpha_k^c A^k \right)$$

where A^k represents feature maps and α_k^c denotes the importance weights derived from gradients. The resulting heatmaps were overlaid on ultrasound images to highlight areas of model attention. This approach enables visual interpretation of the features contributing to classification decisions.

Qualitative analysis showed that correctly classified malignant cases exhibited concentrated activation over lesion regions, indicating that the model focuses on diagnostically relevant structures. In contrast, benign cases typically showed more diffuse or low-intensity activation patterns. Misclassified samples occasionally exhibited spatially shifted or ambiguous attention regions, suggesting potential sources of prediction error. This qualitative assessment provides insight into model behavior and supports interpretability without relying on quantitative localization metrics.

3.05 Research Questions

This study addresses the following research questions:

RQ1:

Can a transfer learning-based ResNet-18 model accurately classify malignant breast ultrasound images?

RQ2:

Does a specificity-driven evaluation framework provide clinically meaningful assessment in diagnostic settings?

RQ3:

Do Grad-CAM visualizations highlight image regions consistent with lesion locations identified by expert annotations?

3.06 Research Hypotheses

The following hypotheses guide this research:

H1:

A transfer learning-based ResNet-18 model can achieve diagnostic accuracy above 90%.

H2:

Model optimization improves specificity by reducing false positive predictions.

H3:

Grad-CAM heatmaps demonstrate meaningful localization over lesion regions in correctly classified malignant cases.

CHAPTER 4 – RESULTS AND DISCUSSION

4.01 Training Performance Analysis

The ResNet-18 transfer learning model was trained for 12 epochs using the BUSI breast ultrasound dataset (Al-Dhabyani et al., 2020). Training accuracy improved from approximately 80% to 98%, while validation accuracy stabilized between 84% and 92%, indicating effective feature learning and reasonable generalization to unseen data.

Training loss decreased consistently across epochs, while validation loss exhibited moderate fluctuations. Such behavior is common in medical imaging tasks due to limited dataset size, class imbalance, and variability in ultrasound image characteristics (Litjens et al., 2017).

The observed gap between training and validation performance suggests mild overfitting, likely influenced by the limited dataset and variability in lesion appearance. However, the absence of significant divergence between the curves indicates that transfer learning with pretrained architectures, such as ResNet-18, effectively stabilizes optimization and supports generalization (He et al., 2016).

Overall, these results demonstrate that the model achieved stable convergence and learned discriminative features suitable for breast ultrasound classification.

4.02 Performance Metrics

The model achieved strong classification performance, as summarized in Table 2, with an overall accuracy of 94.87%, precision of 93.10%, specificity of 97.7%, and a ROC-AUC of 0.9816.

Table 2. Performance Metrics for Breast Cancer Classification

Metric	Value
Accuracy	94.87%
Precision	93.10%
Sensitivity	87.10%
F1 Score	0.90
ROC-AUC	0.9816

The high ROC-AUC indicates excellent separability between malignant and non-malignant classes. Notably, the model demonstrates high specificity, reflecting strong capability in correctly identifying non-malignant cases and minimizing false positive predictions.

4.03 Confusion Matrix Analysis

The confusion matrix provides further insight into classification performance, as shown in Table 3.

Table 3. Confusion Matrix for Malignant vs Non-Malignant Classification

	Predicted Negatives	Predicted Positives
Non - malignant	84	2
Malignant	4	27

The model correctly identified 84 non-malignant cases (true negatives) and 27 malignant cases (true positives), with only 2 false positives and 4 false negatives. The low number of false positives directly contributes to the high specificity (97.7%), demonstrating effective discrimination between malignant and non-malignant cases.

From a diagnostic perspective, this performance is significant, as reducing false positives helps minimize unnecessary biopsies, clinical workload, and patient anxiety.

4.04 Analysis of Misclassification

Misclassification cases provide important insight into model limitations. The two false positive cases suggest that certain benign lesions exhibit visual characteristics similar to malignant patterns, leading to overestimation of malignancy.

False negative cases, although limited in number, were typically associated with challenging imaging conditions. These include low contrast lesions, small lesion size, and irregular or diffuse boundaries, which can obscure distinguishing features in ultrasound images. Difficult samples were often characterized by:

- Low image quality or noise artifacts.
- Weak echogenic contrast between lesion and surrounding tissue.
- Overlapping visual features between benign and malignant cases

These findings highlight the inherent challenges of ultrasound imaging and the limitations of purely pixel-based learning approaches.

4.05 Imaging Limitations and Clinical Implications

Ultrasound imaging presents inherent challenges, including speckle noise, operator dependency, and low contrast resolution compared to other modalities such as CT or MRI. These factors directly affect both human interpretation and automated model performance.

In diagnostic applications, minimizing false positive predictions is particularly important, as unnecessary biopsies and interventions carry both clinical and economic consequences. The model's high specificity demonstrates its effectiveness in reducing such outcomes, supporting its potential role as a decision-support tool.

4.06 Explainability and Clinical Alignment

Grad-CAM visualizations provide insight into model decision-making. In correctly classified cases, activation maps show concentrated attention over lesion regions, aligning with expert-annotated masks and indicating that the model is learning clinically meaningful features.

In misclassified cases, activation patterns are often incomplete or spatially shifted, suggesting that the model fails to fully capture subtle lesion characteristics. This reinforces the importance of explainability in identifying model limitations and guiding future improvements.

4.07 Summary of Findings

The ResNet-18 transfer learning model achieved strong performance, with 94.87% accuracy and a ROC-AUC of 0.9816. The model demonstrated high specificity (97.7%), indicating reliable identification of non-malignant cases and reduced false positive rates.

These results suggest that explainable deep learning models can provide accurate and interpretable diagnostic support in breast ultrasound imaging. While some misclassification challenges remain, particularly in complex imaging conditions, the integration of specificity-driven evaluation and explainability contributes toward clinically meaningful and deployable AI systems.

CHAPTER 5 – CONCLUSION AND FUTURE WORK

5.01 Research Conclusions

This research contributes to the application of artificial intelligence in medical imaging by developing an explainable deep learning framework for breast ultrasound classification. A transfer learning approach based on ResNet-18 was employed, enabling effective feature learning in a limited-data setting.

The model achieved strong diagnostic performance on the BUSI dataset, with high accuracy and a ROC-AUC of 0.9816, demonstrating reliable discrimination between malignant and non-malignant cases. Notably, the model achieved high specificity (97.7%), indicating strong capability in correctly identifying non-malignant cases and reducing false positive predictions in a diagnostic context.

Grad-CAM was integrated to provide visual explanations of model predictions. Qualitative comparison of activation maps with expert-annotated lesion masks showed meaningful localization over lesion regions in correctly classified cases, supporting interpretability and clinical relevance.

Together, these findings demonstrate that combining deep learning with explainable AI can enhance transparency, reliability, and clinical applicability in breast ultrasound imaging.

5.02 Limitations

Despite promising results, several limitations remain. The dataset size is relatively small, which may limit generalization to larger and more diverse clinical populations (Litjens et al., 2017). In addition, the images were collected under controlled conditions, and real-world variability may affect model performance.

Furthermore, Grad-CAM provides qualitative explanations only. The absence of quantitative localization evaluation limits the ability to rigorously assess alignment between model attention and ground-truth pathology.

5.03 Future Work

Future work can extend this research in several directions. Evaluating the model on larger, multi-institutional datasets would improve robustness and generalizability.

Further improvements in classification performance can be explored through advanced data augmentation strategies, class balancing techniques, and architectural enhancements. Multimodal approaches that integrate ultrasound with other imaging modalities, such as mammography or MRI, may further enhance diagnostic capability.

In addition, incorporating advanced explainability techniques, such as SHAP or attention-based models, may provide deeper insights into model decision-making. The proposed framework can also be extended into clinical decision-support systems to assist radiologists in real-time diagnosis (Doi, 2007).

5.04 Final Remarks

This study demonstrates that explainable convolutional neural networks can achieve strong diagnostic performance in breast ultrasound cancer detection while providing interpretable visual insights. The integration of deep learning and explainable AI offers a promising approach for improving diagnostic reliability and supporting clinical decision-making in medical imaging.

References

Al-Dhabyani, W., Gomaa, M., Khaled, H., & Fahmy, A. (2020). Dataset of breast ultrasound images. *Data in Brief*, 28, 104863. <https://doi.org/10.1016/j.dib.2019.104863>

American Cancer Society. (2023). *Breast cancer facts & figures 2023–2024*. <https://www.cancer.org>

Doi, K. (2007). Computer-aided diagnosis in medical imaging: Historical review, current status and future potential.

Computerized Medical Imaging and Graphics, 31(4–5), 198–211.

<https://doi.org/10.1016/j.compmedimag.2007.02.002>

Esteva, A., Kuprel, B., Novoa, R. A., Ko, J., Swetter, S. M., Blau, H. M., & Thrun, S. (2017). Dermatologist-level classification of skin cancer with deep neural networks. *Nature*, 542(7639), 115–118.

<https://doi.org/10.1038/nature21056>

He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)* (pp. 770–778).

Kopans, D. B. (2007). *Breast imaging* (3rd ed.). Lippincott Williams & Wilkins.

Litjens, G., Kooi, T., Bejnordi, B. E., Setio, A. A. A., Ciompi, F., Ghafoorian, M., van der Laak, J. A. W. M., van Ginneken, B., & Sánchez, C. I. (2017). A survey on deep learning in medical image analysis. *Medical Image Analysis*, 42, 60–88.

<https://doi.org/10.1016/j.media.2017.07.005>

National Cancer Institute (NCI). (2023). *Cancer stat facts: Female breast cancer*.

<https://seer.cancer.gov/statfacts/html/breast.html>

Parikh, R. B., Teeple, S., & Navathe, A. S. (2019). Addressing bias in artificial intelligence in health care. *JAMA*, 322(24), 2377–2378. <https://doi.org/10.1001/jama.2019.18058>

Rajpurkar, P., Irvin, J., Zhu, K., Yang, B., Mehta, H., Duan, T., Ding, D., Bagul, A., Langlotz, C., Shpanskaya, K., Lungren, M. P., & Ng, A. Y. (2018). CheXNet: Radiologist-level pneumonia detection on chest X-rays with deep learning. *arXiv preprint arXiv:1711.05225*.

- Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., & Batra, D. (2017). Grad-CAM: Visual explanations from deep networks via gradient-based localization. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)* (pp. 618–626).
- Skaane, P. (2017). Breast cancer screening with digital breast tomosynthesis. *Breast Cancer Research*, 19(1), 1–9.
<https://doi.org/10.1186/s13058-017-0859-3>
- World Health Organization (WHO). (2023). *Cancer fact sheet*. <https://www.who.int/news-room/fact-sheets/detail/cancer>
- Frimpong, S. K. (2026). *BUSI explainable CNN with Grad-CAM* [Computer software]. GitHub.
https://github.com/frimpongsarfok/AI_Breast_Cancer_Classification_VSU/blob/main/BUSI_Explainable_CNN_GradCAM_Colab.ipynb

